

Equations | practice and reasoning

1. Solve $\cos(x) = -\frac{1}{2}$ for $0 \leq x < 360^\circ$.
2. Solve $\log_{10}(4x + 968) = 3$. State the domain restriction.
3. Find the intersection of $y = -3x - 9$ and $y = 4x + 19$. Show your working.
4. Find the roots of $\frac{1}{2}x^2 - x = 0$. Give exact answers and show your working.

Solutions

1. $x = 120^\circ, 240^\circ$. Use the unit circle; 360 degrees is excluded.

2. Domain: $4x + 968 > 0$.

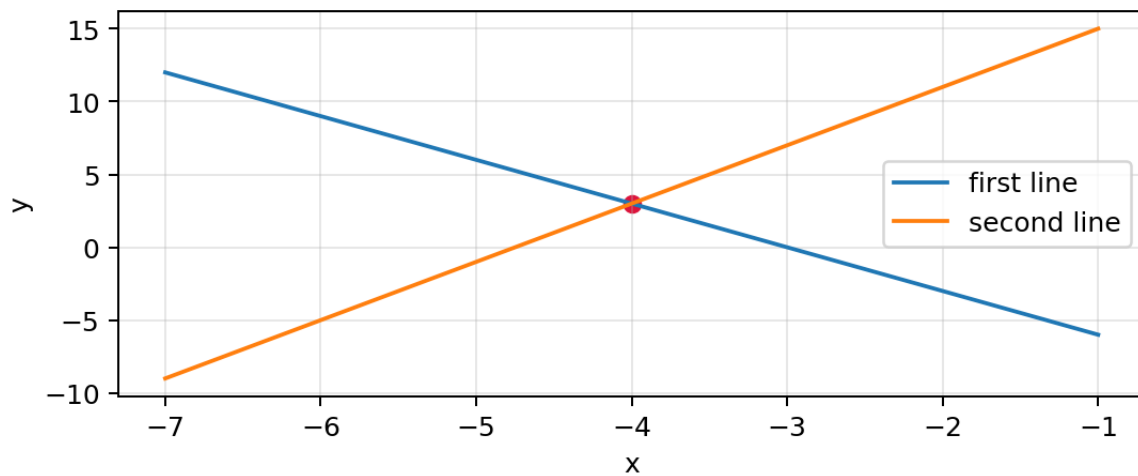
Rewrite: $4x + 968 = 10^3 = 1000$.

$x = 8$. Check: the logarithm argument is $1000 > 0$.

3. Set the expressions equal: $-7x = 28$.

$x = -4$; substitute into either equation to get $y = 3$.

Intersection: $(-4, 3)$.



4. Factor: $\frac{x}{2}(x - 2) = 0$. Therefore $x = 0, 2$.